FoodHub Order Analysis Low-Code Project

Methodology and Workflow

Exploratory Data Analysis and Business Intelligence for a New York City Food Aggregator

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1. Executive Summary

The **FoodHub** food aggregator company, located in **New York City**, operates in a densely populated area that offers key advantages, including a wide range of **Restaurants**, high customer demand, and substantial **Weekend Order** volume. FoodHub operates using a smartphone app that allows customers to place direct orders to a wide range of restaurants, while also using designated delivery people to pick up the **Order** from a restaurant and drop it off at the customer's location. Customers can then leave a **Rating** based on the service they received. The FoodHub company earns a fixed-margin revenue from delivery orders placed by restaurants.

Key Conclusions

- Strong **Weekend** demand driven by a strong inclination toward **American cuisine** among the top-performing Restaurants.
- The company should focus on developing strong partnerships with top-performing Restaurants (e.g., **Shake Shack** and **The Meatball Shop**).
- **Delivery Times** are efficient, averaging less than 30 minutes.
- Many **Ratings** are missing, causing inaccurate customer satisfaction analysis.
- Strong baseline of positive experiences, driven by an abundance of **5-star Ratings**.

- Current **Total Net Revenue** from the dataset is approximately **$6,166.30**, where most orders do not surpass $20.

Key Actionable Insights

- 20 top-performing Restaurants responsible for high order count and Revenue.
- Delivery Times on **Weekends (22 minutes)** are faster than on **Weekdays (28 minutes)**.
- Majority of orders (**89.46%**) do not take longer than 60 minutes to deliver.
- Most customers order **American**, **Japanese**, and **Italian** cuisines.
- Customer Ratings drop when delivery times exceed ~25 minutes, suggesting delivery time strongly correlates with customer satisfaction.
- Restaurants with high Ratings (**The Meatball Shop**, **Blue Ribbon Fried Chicken**, **Shake Shack**) are preferred for promotional campaigns.

Key Recommendations

- Launch loyalty programs and upsell bundling.
- Promote top-performing Restaurants through marketing campaigns.
- Emphasize Weekends given higher demand.
- Create premium offers and subscription plans; partner with high-performing Restaurants.
- Implement in-app prompts and push notifications to reduce "Not given" Ratings.
- Expand kitchen staffing on Weekdays to minimize delays.
- Implement an optimized navigation algorithm for delivery routes.

2. Business Problem Overview and Solution Approach

2.1 Business Problem Overview

FoodHub connects customers to a multitude of **Restaurants** across New York City via a smartphone app. Despite growing demand for online delivery among customers — particularly students and busy professionals — the company faces several critical challenges:

- **Customer retention and engagement** — Many customers fail to return and do not leave Ratings, creating blind spots in satisfaction measurement.
- **Inconsistencies in Delivery Times** — Approximately 1 in 10 (10%) orders exceed 60 minutes total delivery time.
- **Commission fee disapproval** — Roughly 30% of partnering Restaurants express dissatisfaction with the commission fee structure.
- **Revenue optimization** — Fewer than one-third of all orders exceed $20, limiting higher-tier commission revenue.
- **Revenue concentration** — A small number of top-performing Restaurants account for the majority of Revenue, creating dependency risk.

2.2 Solution Approach

FoodHub follows a **data-driven, multi-faceted methodology** using Python-driven statistical modeling and computations conducted by the Data Science team. The approach emphasizes Python-powered statistical analyses (code snippets referenced in the appendix). The key investigation areas include:

- **Customer clustering behavioral metrics** — Identifying top customers and cuisine preferences to drive targeted retention strategies.
- **Restaurant performance benchmarking** — Comparing Number of Orders, Revenue, and Ratings across all partnering Restaurants.
- **Operational metrics** — Tracking Food Preparation and Delivery Times to identify bottlenecks and improvement opportunities.
- **Delivering projections** — Addressing customer retention, delivery efficiency, Restaurant partnering satisfaction, and revenue optimization through quantitative forecasting.

3. Data Overview

3.1 Dataset Description

The dataset collected by FoodHub includes **1,898 unique Orders** (each represented by a row) and **9 columns**. Rigorous data quality checks confirmed **no missing values** and **no duplicate entries** were found, establishing a clean foundation for analysis.

3.2 Categorical Variables

Variable	Description	Unique Values
Order ID	Unique identifier for each Order	1,898
Customer ID	ID of the customer placing an order	1,200
Restaurant Name	Name of the Restaurant fulfilling the Order	178
Cuisine Type	Type of cuisine being ordered	14
Day of the Week	Weekend (Saturday, Sunday) or Weekday (Monday–Friday)	2

3.3 Numerical Variables

Variable	Description	Data Type
Cost of the Order	Cost of each Order in USD	Float
Rating	Customer Rating after receiving Order (values: "Not given", 5, 4, 3)	Object
Food Preparation Time	Time taken for the Restaurant to prepare food (minutes)	Integer
Delivery Time	Time taken for delivery employee to deliver food (minutes)	Integer

3.4 Data Types

- **Integer types:** Order ID, Customer ID, Food Preparation Time, Delivery Time
- **Float type:** Cost of the Order
- **Object types:** Restaurant Name, Cuisine Type, Day of the Week, Rating

> **Note**
>
> Order ID and Customer ID were later converted to **object** type (identifiers, not numerical values).
>
> Rating was later converted to **integer** for numerical calculations and statistical analysis.

3.5 Summary Descriptive Statistics

Variable	Count	Mean	Std	Min	25%	50%	75%	Max
cost_of_the_order	1898.0	16.50	7.48	4.47	12.08	14.14	22.30	35.41
food_preparation_time	1898.0	27.37	4.63	20.00	23.00	27.00	31.00	35.00
delivery_time	1898.0	24.16	4.97	15.00	20.00	25.00	28.00	33.00

Key observations from descriptive statistics:

- **Cost of the Order:** Average ~$17; mid-range affordable pricing; minimum ~$4.47, maximum ~$35.41.
- **Food Preparation Time:** Average ~27 minutes; suitable pace; range 20–35 minutes.
- **Delivery Time:** Average ~24 minutes; efficient service; most customers wait under 30 minutes.

3.6 Rating Distribution

Rating	Count
Not given	736
5	588
4	386

Rating	Count
3	188

Important Observation

No Ratings of "1" or "2" exist in the dataset. This indicates that unsatisfied customers chose not to rate their experience, or that the majority of customers were mainly satisfied with the service. The absence of low Ratings introduces a **positive bias** in the satisfaction data.

4. Exploratory Data Analysis — Univariate Analysis

The univariate analysis methodology involves the examination of the nature, distribution, and implications of each variable individually. **Bar plots** were used for categorical variables, while **histograms** and **boxplots** were used for numerical variables. This approach establishes the baseline understanding of each feature before exploring inter-variable relationships.

4.1 Categorical Variables

Order ID

The dataset contains **1,898 unique Order IDs**, confirming that each Order is unique and matches the total number of rows in the dataset. This validates the data integrity — no duplicate transactions exist.

Customer ID

A total of **1,200 unique customers** are registered in the dataset. Since there are 1,898 orders but only 1,200 unique customers, this confirms that a subset of customers placed multiple orders — indicating repeat patronage and an opportunity for loyalty segmentation.

Restaurant Name

There are **178 unique Restaurants** providing a wide variety of Cuisine Types. Both top-performing and underperforming Restaurants are responsible for varying levels of revenue generation, highlighting the need for targeted partnership strategies.

Cuisine Type (Graph 1)

The dataset features **14 distinct cuisines**. The distribution is highly skewed — a few cuisines dominate while many receive comparatively few orders, evidencing both substantial variety and strong global appeal across the NYC market.

Demand Tier	Cuisines	Approx. Orders
High Demand	American, Japanese, Italian	~580, ~470, ~300
Moderate Demand	Chinese, Mexican, Indian, Mediterranean, Middle Eastern	~220, ~80, ~80, ~50, ~50
Niche	Thai, Southern, French, Spanish, Vietnamese, Korean	~30 each; ~20 each (Vietnamese, Korean)

Day of the Week (Graph 4)

The variable contains two unique values: **Weekend** and **Weekday**. Weekend dominance is evident with approximately **1,400 orders** — about double Weekday volume. The dataset is heavily skewed toward Weekend activity. Higher Weekend demand is possibly driven by leisure time, social gatherings, and dining-out behavior. Lower Weekday demand implies routine work schedules limit ordering frequency.

Rating (Graph 5)

Four unique values are present: **"Not given"**, **"5"**, **"4"**, and **"3"**. "Not given" dominates with 736 occurrences. Rating "5" is the second most common (~588), suggesting a strong high-satisfaction inclination. Rating "4" (~386) indicates moderate satisfaction, while Rating "3" (~188) is the lowest, suggesting neutral or slightly negative satisfaction. A clear **positive bias** is present: Ratings "5" and "4" account for a substantial share of all submitted Ratings. The skewed distribution with fewer "3" ratings reinforces the generally positive customer experience.

4.2 Numerical Variables

Cost of the Order (Graphs 2–3)

- **Histogram:** Slightly right-skewed distribution; peak density in the $10–$15 range; secondary peaks observed at $25 and $30.
- **Boxplot:** Median ~$15, positioned closer to Q1 (~$12); Q3 ~$22; full range $5–$35.
- **Implication:** Opportunities exist to upsell, bundle, and run premium promotions to shift the distribution rightward.

Food Preparation Time (Graphs 6–7)

- **Histogram:** Multimodal distribution with peaks at 20, 26, 30, and 34 minutes, suggesting different cuisines and product types have distinct preparation profiles. Common durations span 22–33 minutes.
- **Boxplot:** Relatively symmetric, slightly right-skewed; Q1 = 23 min, Q3 = 31 min; median = 27 min; range 20–35 min.

Delivery Time (Graphs 8–9)

- **Histogram:** Slightly right-skewed; multi-structured peaks clustering at 25 and 28 minutes; secondary peaks at 15 and 20 minutes; wide spread suggests high variability in delivery logistics.
- **Boxplot:** Median ~25 min; Q1 = 20 min, Q3 = 28 min; range 15–33 min.
- **Implication:** Need to optimize routes, improve hotspot delivery coverage, and enhance customer communication for orders approaching the upper range.

5. Univariate Analysis — Answers to Business Application Research Plans

5.1 Top 5 Restaurants by Number of Orders

Rank	Restaurant Name	Order Count
1	**Shake Shack**	219

Rank	Restaurant Name	Order Count
2	**The Meatball Shop**	132
3	**Blue Ribbon Sushi**	119
4	**Blue Ribbon Fried Chicken**	96
5	**Parm**	68

A notable brand relationship exists between **Blue Ribbon Sushi** (Japanese cuisine) and **Blue Ribbon Fried Chicken** (American cuisine), demonstrating cross-cuisine brand loyalty within the same restaurant group.

5.2 Most Popular Cuisine Type on Weekends

American cuisine leads with **415 orders** on Weekends — reflecting a strong customer inclination toward familiar comfort food during leisure time. American cuisine accounts for the majority of Weekend Revenue and is a primary driver of the overall Weekend demand spike.

5.3 Percentage of Orders Above $20

- Total orders above $20: **555**
- Percentage of total: **29.24%**
- Conversely, **70.76%** of orders cost $20 or less
- Approximately **1 in 3 orders** qualifies as a high-value order

This distribution highlights the opportunity for marketing strategies focused on upselling and bundling to shift more orders above the $20 threshold, where the 25% commission tier generates substantially higher net revenue.

5.4 Mean Order Delivery Time

The mean delivery time across all orders is **24.16 minutes**. Customers on average do not wait more than 30 minutes, which is a typical and acceptable delivery window for food aggregator platforms operating in dense urban environments like New York City.

5.5 Top 3 Most Frequent Customers (Eligible for 20% Discount Vouchers)

Rank	Customer ID	Order Count
1	**52832**	13
2	**47440**	10
3	**83287**	9

These are highly engaged, loyal, and satisfied customers who contribute disproportionately to overall Revenue. They are prime candidates for the **20% discount voucher program** to reinforce retention and encourage continued patronage.

6. Exploratory Data Analysis — Bivariate Analysis

The multivariate analysis methodology examines the relationships between numerical and categorical variables using **boxplots**, **pointplots**, **heatmaps**, and **cross-tabulations**. This phase identifies patterns, dependencies, and operational benchmarks that inform targeted business strategies.

6.1 Cuisine Type vs. Cost of the Order (Graph 10)

Cuisine types were organized by cost tier based on median order cost and interquartile range:

Higher-Cost Tier

- **Southern:** Median ~$15, range ~$7–$31
- **Japanese:** Median ~$15, range ~$5–$34
- **French:** Median ~$22, range ~$13–$30

Premium options, specialty dishes, or bundled orders characterize this tier.

Moderate-Cost Tier

- **Italian:** Median ~$15, Q1 $13, Q3 $21
- **Mediterranean:** Median ~$14, Q1 $13, Q3 $18
- **American:** Median ~$14, Q1 $13, Q3 $22
- **Mexican:** Median ~$16, Q1 $13, Q3 $23
- **Indian:** Median ~$16, Q1 $12, Q3 $24
- **Spanish:** Median ~$16, Q1 $14, Q3 $24

This tier demonstrates balanced and consistent pricing across products.

Lower-Cost Tier

- **Korean:** Median ~$13, Q1 $12, Q3 $14
- **Vietnamese:** Median ~$11, Q1 $10, Q3 $12
- **Middle Eastern:** Median ~$17, Q1 $12, Q3 $24
- **Chinese:** Median ~$15, Q1 $14, Q3 $22
- **Thai:** Median ~$19, Q1 $13, Q3 $24

Korean and Vietnamese have low medians and narrow interquartile ranges — consistent, affordable options. Outlier data points were noted for Korean (3 below lower bound, 2 above max), Mediterranean (several above upper bound), and Vietnamese (1 above upper bound).

Insight

Most customers prefer lower-cost cuisines. Marketing should focus on both premium and budget cuisine types to maximize reach across customer segments.

6.2 Cuisine Type vs. Food Preparation Time (Graph 11)

Cuisine types were categorized by preparation speed to identify operational efficiency patterns:

Faster Preparation Tier

- **Korean:** Median ~25 min, Q1 23, Q3 26, range 20–29
- **Vietnamese:** Median ~25 min, Q1 22, Q3 29, range 20–32
- **Middle Eastern:** Median ~27 min, Q1 24, Q3 30, range 20–34
- **Chinese:** Median ~27 min, Q1 23, Q3 32, range 20–35

Moderate Preparation Tier

- **Mexican:** Median ~26 min, Q1 23, Q3 31, range 20–35
- **American:** Median ~27 min, Q1 23, Q3 31, range 20–35
- **Indian:** Median ~27 min, Q1 23, Q3 31, range 20–35
- **Mediterranean:** Median ~27 min, Q1 23, Q3 31, range 20–35
- **Spanish:** Median ~28 min, Q1 24, Q3 29, range 20–35
- **Italian:** Median ~28 min, Q1 23, Q3 31, range 20–35

Longest Preparation Tier

- **Japanese:** Median ~28 min, Q1 24, Q3 32, range 20–35
- **French:** Median ~24 min, Q1 23, Q3 31, range 21–35
- **Thai:** Median ~28 min, Q1 22, Q3 32, range 21–35
- **Southern:** Median ~26 min, Q1 24, Q3 32, range 20–35

Two outlier data points were observed for Korean cuisine above the upper bound. The operational recommendation is to promote faster-prep cuisines through quick-delivery campaigns while streamlining long-prep cuisines through menu and staffing optimization.

6.3 Day of the Week vs. Delivery Time (Graph 12)

Metric	Weekend	Weekday
Median Delivery Time	~22.5 min	~27.5 min
Q1	~18 min	~27.5 min
Q3	~27.5 min	~30 min

Metric	Weekend	Weekday
Range	15–30 min	25–32.5 min

Weekend deliveries are quicker, likely due to increased staffing or more equitable demand distribution. **Weekday** deliveries are longer but more consistent; traffic from workday congestion contributes to delays. The recommendation is to implement more Weekday delivery staff and promote faster Weekend delivery times to attract additional customers.

6.4 Rating vs. Delivery Time (Graph 13 — Pointplot)

Rating	Avg. Delivery Time	Variability
Not given	Below ~24.25 min	Shorter confidence interval — consistent
3	~24.5 min (slowest)	Longer confidence interval — high variability
4	Below 24 min (fastest)	Shorter confidence interval — consistent
5	Slightly below ~24.5 min	Shorter confidence interval — consistent

Operational benchmark: Keep delivery under 25 minutes to retain customer satisfaction and minimize low-Rating occurrences.

6.5 Rating vs. Food Preparation Time (Graph 14 — Pointplot)

- **"Not given":** Average slightly below 27.4 min
- **Rating "3":** Slowest average (~27.4 min) — lower satisfaction linked to longer prep times
- **Rating "4":** Similar average (~27.4 min)
- **Rating "5":** Fastest average (slightly below 27.4 min)

Service quality target: Keep food preparation below 27 minutes. Menu staffing review should be linked to cuisines consistently receiving lower Ratings.

6.6 Rating vs. Cost of the Order (Graph 15 — Pointplot)

- **"Not given":** Lowest average cost (~$16) — lighter, smaller orders

- **Rating "3":** ~$16.50 — medium-priced orders do not always lead to higher satisfaction
- **Rating "4":** ~$16.75 — more expensive orders correlate with higher satisfaction
- **Rating "5":** Highest average (~$17) — customers spending more tend to be more satisfied

This positive correlation between order cost and satisfaction suggests that a pricing strategy guaranteeing a premium experience could reinforce satisfaction. Service quality should be aligned with price tiers to meet customer expectations.

6.7 Correlation Heatmap (Graph 16)

	cost_of_the_order	food_preparation_time	delivery_time
cost_of_the_order	1.00	0.04	−0.03
food_preparation_time	0.04	1.00	0.01
delivery_time	−0.03	0.01	1.00

Interpretation of correlations:

- **Cost vs. Food Prep (+0.04):** Slight positive correlation; more expensive orders take marginally longer to prepare, but the effect is imperceptible.
- **Cost vs. Delivery Time (−0.03):** Slight negative correlation; expensive orders may be delivered slightly faster, but the relationship is inconsequential.
- **Food Prep vs. Delivery Time (+0.01):** Non-meaningful relationship; the two operational processes are effectively independent.

Analytical Implication

All correlations are extremely weak. Each variable is influenced by independent operational factors. Predictive models should not assume strong interdependence among these features — they should be treated as separate features for optimization purposes.

7. Bivariate Analysis — Answers to Business Application Research Plans

7.1 Restaurants Eligible for Promotional Offer

Top 5 Restaurants by Total Rating Count:

Restaurant Name	Rating Count
Shake Shack	133
The Meatball Shop	84
Blue Ribbon Sushi	73
Blue Ribbon Fried Chicken	64
RedFarm Broadway	41

Restaurants eligible for Promotional Offer (Rating count >50 AND average rating >4):

Restaurant Name	Average Rating
The Meatball Shop	4.51
Blue Ribbon Fried Chicken	4.33
Shake Shack	4.28
Blue Ribbon Sushi	4.22

The Meatball Shop has the highest average rating (4.51) and stands out as the top candidate for promotional campaigns. The **Blue Ribbon** brand relationship across cuisines (Sushi and Fried Chicken) demonstrates cross-cuisine brand loyalty. Notably, The Meatball Shop and Blue Ribbon Fried Chicken achieve higher ratings than Shake Shack despite having fewer total orders — suggesting superior quality-per-order delivery.

7.2 Revenue Generated Across Top 14 Performing Restaurants

Rank	Restaurant Name	Revenue ($)
1	**Shake Shack**	3,579.53
2	**The Meatball Shop**	2,145.21
3	**Blue Ribbon Sushi**	1,903.95
4	**Blue Ribbon Fried Chicken**	1,662.29
5	Parm	1,112.76
6	RedFarm Broadway	965.13
7	RedFarm Hudson	921.21
8	TAO	834.50
9	Han Dynasty	755.29
10	Blue Ribbon Sushi Bar & Grill	666.62
11	Rubirosa	660.45
12	Sushi of Gari 46	640.87
13	Nobu Next Door	623.67
14	Five Guys Burgers and Fries	506.47

Shake Shack is the top revenue performer at $3,579.53 — approximately **67% higher** than the second-place restaurant. Strong branding clusters are evident: the **Blue Ribbon** family (Sushi, Fried Chicken, Sushi Bar & Grill) and **RedFarm** (Broadway and Hudson) each appear multiple times, demonstrating brand strength across locations. Revenue dependency on a small number of restaurants represents both a risk and an opportunity for growth through mid-tier restaurant development.

7.3 Total Net Revenue

FoodHub operates on a **tiered commission revenue model**:

Order Value Tier	Commission Rate
Orders > $20	25%
Orders $5–$20	15%
Orders < $5	0%

Total Net Revenue: ~$6,166.30. The majority of Revenue is generated from the **15% commission tier**, since the bulk of orders cluster in the $10–$15 range. Shifting more orders above the $20 threshold through upselling and bundling would significantly increase net revenue per order.

7.4 Percentage of Orders Exceeding 60 Minutes Total Delivery Time

- Orders exceeding 60 minutes (Food Preparation + Delivery Time combined): **200**
- Percentage of total: **10.54%**
- Nearly **90% of orders** are delivered in under 60 minutes

The 10.54% of delayed orders is likely attributable to complicated food preparation, peak demand periods, traffic congestion, and longer delivery distances. Targeted interventions should focus on reducing this percentage to below 5%.

7.5 Mean Delivery Times — Weekdays vs. Weekends

Day Type	Mean Delivery Time	Difference
Weekday	~28 minutes	6 minutes (21% faster on Weekends)
Weekend	~22 minutes	

Inferences: Weekend deliveries are **21% faster** than Weekday deliveries, with a 6-minute difference. Contributing factors include less traffic on Weekends, restaurants may have more staff on Weekends, and smaller Weekend orders may lead to shorter preparation times. Aligning Weekday delivery performance closer to Weekend levels is a key operational priority.

8. Conclusions and Recommendations

8.1 Key Conclusions

1. **Top frequent customers drive disproportionately more Revenue;** identify and reward the top 20 customers through loyalty programs.
2. **Weekend demand is stronger than Weekday;** promotions and upselling strategies should emphasize Weekends as the primary revenue window.
3. **A Ratings gap exists** — the majority of Ratings are "Not given"; incentivizing customers to leave Ratings is critical for accurate satisfaction analysis.
4. **Revenue is concentrated in three main outlets:** Shake Shack, The Meatball Shop, and Blue Ribbon, creating dependency risk.
5. **Cross-brand loyalty** is evident from familiar brand names (Blue Ribbon Sushi/Fried Chicken; RedFarm Broadway/Hudson).
6. **Delivery Times are longer on Weekdays;** cuisines like Japanese, French, Thai, and Southern have longer Food Preparation Times.
7. **American cuisine is most preferred on Weekends,** representing the largest single revenue category.
8. **Tiered net revenue commissions** (25% for orders >$20; 15% for orders $5–$20) mean that the majority of order costs falling below $20 limits revenue potential.
9. **Cost, Food Prep Time, and Delivery Time have weak correlations** — each should be optimized independently through targeted operational strategies.

8.2 Recommendations

Marketing and Revenue

- Feature high-Rated Restaurants in promotional campaigns to drive brand awareness and order volume.
- Promote **American cuisine on Weekends** to capitalize on existing demand and increase Revenue.
- Feature Japanese and Italian as **premium advertising targets**; Vietnamese and Korean as **affordable alternatives**.

- Implement **upselling and bundling** strategies to increase average order value past the $20 threshold.
- Deploy **loyalty rewards and exclusive deals** for top customers to reinforce retention.
- Create **combo deals in the $12–$15 range**; develop premium food pricing for high-paying customer segments.

Operations and Logistics

- Align **Weekday delivery performance** closer to Weekend delivery times through staffing optimization.
- Standardize food preparation workflows for **longer-prep-time Cuisine Types** to reduce variability.
- **Expand kitchen staffing on Weekdays** to minimize delays and improve throughput.
- Implement an **optimized navigation algorithm** for delivery routes to reduce delivery times.
- **Consolidate high-demand delivery locations** to improve logistics efficiency and reduce costs.

Customer Experience

- Implement **in-app prompts and push notifications** to encourage customers to leave Ratings after each order.
- Provide **incentives** (discounts, loyalty points) for customers who leave Ratings to reduce the "Not given" proportion.
- Deploy **follow-up surveys** to capture structured feedback and reduce the Ratings gap.
- Ensure sufficient **delivery drivers and food prep staff** during peak demand windows.

Strategic Positioning

- Reinforce **customer retention and acquisition** across platforms through multi-channel engagement.
- Incorporate **business continuity capabilities** by scaling logistics and technology infrastructure.
- **Partner strongly with top-performing Restaurants** to secure exclusive deals and preferential placement.

- Differentiate FoodHub as a **quality-first aggregator** with a curated marketplace of high-performing restaurant partners.
- Use **predictive techniques** based on day type, Cuisine Type, and Delivery Time for demand forecasting and resource allocation.

9. Workflow Summary

The following numbered workflow synthesizes the complete end-to-end methodology applied in this FoodHub Low Code Project:

10. **Define the business problem** — FoodHub customer retention, delivery efficiency, and revenue optimization in New York City.
11. **Establish the solution approach** — Python-driven EDA, customer clustering, restaurant benchmarking, and operational metrics analysis.
12. **Acquire the FoodHub dataset** — 1,898 orders across 9 columns representing customer, restaurant, and operational data.
13. **Conduct data quality checks** — Verify missing values, duplicates, and data types; confirm data integrity.
14. **Convert data types as needed** — Order ID and Customer ID to object (identifiers); Rating to integer for numerical calculations.
15. **Compute summary descriptive statistics** — Generate mean, standard deviation, min, max, and quartile values for numerical variables.
16. **Analyze Rating distribution** — Identify the "Not given" gap and positive bias in customer Ratings.
17. **Perform univariate analysis on categorical variables** — Bar plots for Cuisine Type, Day of Week, and Rating distributions.
18. **Perform univariate analysis on numerical variables** — Histograms and boxplots for Cost of the Order, Food Preparation Time, and Delivery Time.

19. **Answer business research questions** — Top 5 Restaurants, most popular Weekend cuisine, percentage of orders >$20, mean delivery time, and top 3 frequent customers.
20. **Perform multivariate analysis** — Cuisine Type vs. Cost (boxplots), Cuisine Type vs. Food Prep Time (boxplots), Day of Week vs. Delivery Time (boxplots).
21. **Analyze Rating relationships** — Rating vs. Delivery Time (pointplot), Rating vs. Food Prep Time (pointplot), Rating vs. Cost (pointplot).
22. **Compute correlation heatmap** — Assess interdependencies among cost_of_the_order, food_preparation_time, and delivery_time.
23. **Answer multivariate business research questions** — Promotional offer eligibility, top 14 revenue restaurants, total net revenue, percentage of orders >60 min, and mean weekday vs. weekend delivery times.
24. **Compute tiered revenue model** — 25% commission for orders >$20, 15% for orders $5–$20, and 0% for orders <$5.
25. **Derive actionable insights** — Synthesize findings across customer behavior, restaurant performance, and operations.
26. **Formulate strategic recommendations** — Organize into marketing, operations, customer experience, and strategic positioning categories.
27. **Communicate findings to stakeholders** — Present all conclusions with quantitative rationale and data-backed justification.